

# Internship Research Brief

Data Analyst Platform • Sumvec.AI • April – May 2026

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| <b>Intern</b>      | Akanksha Parab, St. Xavier's College                                                |
| <b>Degree</b>      | B.Sc. Mathematics (Major), Statistics (Minor) — Year II                             |
| <b>Mentor</b>      | Sasi Maadavan Ramesh Babu, Sumvec.AI                                                |
| <b>Duration</b>    | 1 April 2026 – 31 May 2026 (120 hours, as per NEP guidelines)                       |
| <b>Paper Title</b> | Analysing Query Embedding Spaces in RAG-Based Text-to-SQL Systems Using PCA and SVM |

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## Background

### The Data Analyst Platform

Sumvec's Data Analyst platform is a production Retrieval-Augmented Generation (RAG) system that enables non-technical users to query enterprise databases using natural language. The platform converts plain-English questions into SQL queries, executes them, and returns structured results or visualisations.

At the core of the retrieval mechanism, each natural language query is encoded as a high-dimensional vector (embedding) using the BAAI/bge-small-en-v1.5 model, producing 384-dimensional representations. These embeddings are stored in a Pinecone vector database. When a new query arrives, the system retrieves the most semantically similar past queries to assist SQL generation via a Claude LLM.

While the platform is operational and performant, the geometric properties of the embedding space — how well queries cluster by type and complexity, how much of the 384 dimensions carry meaningful signal, and whether query complexity can be inferred from embeddings alone — remain uncharacterised. This research addresses that gap.

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## Problem Statement

RAG-based Text-to-SQL systems rely on embedding similarity to retrieve relevant examples before generating SQL. The quality of retrieval is therefore directly tied to the structure of the embedding space. However, several questions about this space remain open:

- How much of the variance in 384-dimensional query embeddings is captured by a small number of principal components? Is the effective dimensionality of the space significantly lower than 384?
- Do queries of similar syntactic or semantic complexity cluster together in the embedding space, or are they geometrically dispersed?
- Can a classifier trained on dimensionality-reduced embeddings reliably distinguish query types (e.g. simple projection, aggregation, multi-table join, nested subquery)?

- What is the sensitivity of classification accuracy to the number of PCA components retained? At what point does further dimensionality reduction degrade downstream performance?

Answers to these questions have direct implications for the Data Analyst platform: they inform whether query routing by complexity is feasible, whether the current embedding model is well-suited to the task, and whether the Pinecone index can be optimised.

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## Research Objectives

The internship will pursue the following primary objectives, with stretch goals defined separately.

### Primary Objectives

- **Objective 1 — PCA Variant Selection:** Identify and justify the most appropriate variant of PCA for this problem context. This includes evaluating standard PCA, Kernel PCA, Sparse PCA, and Incremental PCA against the properties of high-dimensional text embeddings — considering linearity assumptions, computational feasibility, and interpretability of components.
- **Objective 2 — Embedding Space Characterisation:** Apply the selected PCA variant to a large corpus of natural language database queries to quantify variance explained per component and identify the effective intrinsic dimensionality of the space.
- **Objective 3 — Cluster Analysis:** Evaluate whether queries of similar type and complexity form coherent geometric clusters in PCA-reduced and t-SNE-reduced representations.

### Stretch Goals

The following objectives are to be pursued if primary objectives are completed ahead of schedule.

- **Stretch Goal 1 — Query Type Classification:** Train and evaluate an SVM classifier on PCA-reduced embeddings to distinguish query complexity categories. Assess precision, recall, and F1 score across categories.
  - **Stretch Goal 2 — Sensitivity Analysis:** Characterise the relationship between the number of retained PCA components and downstream SVM classification accuracy. Identify the optimal dimensionality trade-off.
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## Scope and Dataset

### Dataset

The research will use the Spider dataset, a publicly available cross-domain Text-to-SQL benchmark comprising over 10,000 annotated natural language question–SQL pairs across 200 databases. Spider is a standard academic benchmark in the Text-to-SQL literature and

contains queries spanning a wide range of SQL complexity levels (Easy, Medium, Hard, Extra Hard), making it well-suited to this analysis.

No proprietary data from Sumvec's production systems will be used. All experiments are reproducible from public sources.

## Embedding Model

Embeddings will be generated using **BAAI/bge-small-en-v1.5** via the Sentence-Transformers library — the same model in production use on the Data Analyst platform. This ensures the findings are directly applicable to the existing system.

## Methods

- **Dimensionality Reduction:** PCA variant selected through Objective 1, alongside t-SNE, implemented via scikit-learn.
- **Classification (Stretch):** Support Vector Machine (SVM) with comparative evaluation across linear, RBF, and polynomial kernels.
- **Sensitivity Analysis (Stretch):** Systematic variation of PCA component count against SVM accuracy, with identification of the elbow point.
- **Statistical Evaluation:** Variance explained ratios, reconstruction error, cluster cohesion metrics, and — where stretch goals are reached — precision, recall, F1 score, and confusion matrices.

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## Expected Deliverables

- **Research Paper:** A complete paper submitted to arXiv (cs.IR / cs.LG) by 31 May 2026, authored by Akanksha Parab.
- **Codebase:** A reproducible Python codebase (Jupyter notebooks) covering embedding generation, PCA variant analysis, cluster visualisation, and — if applicable — SVM training and sensitivity analysis.
- **Findings Report:** A summary of key findings and their practical implications for the Data Analyst platform, to be presented internally at Sumvec.

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## Relevance to Sumvec

Beyond the academic contribution, the research is expected to produce actionable insights for the Data Analyst platform:

- The PCA variant selection analysis will establish a principled basis for any future dimensionality reduction applied to the Pinecone embedding index.
- If effective dimensionality is substantially lower than 384, there is a case for using a smaller embedding model or truncating dimensions — reducing Pinecone storage and retrieval latency.

- If the SVM classifier achieves reliable query-type accuracy (stretch goal), it provides the technical foundation for a query routing layer — directing simple queries to faster, lower-cost models and complex queries to more capable ones.
  - Cluster analysis of the embedding space will reveal which query types are poorly represented in the current training data, directly informing Pinecone index improvements.
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Prepared by

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